

Jaewan Park

📍 Champaign, IL, USA ✉ jaewanp2@illinois.edu 🌐 Website in Jaewan Park G Google Scholar

Research Interests

Agentic AI, Foundation Models, and Neural Operators for Scientific Discovery

- **Towards Autonomous AI Scientists:** Training RL-driven LLM agents capable of hypothesis generation, verification, and zero-shot synthesis of complex PDE solvers.
- **Efficient Adaptation of Foundation Models:** Designing model-agnostic frameworks for downstream tasks and efficient finetuning that transfer across diverse domains, with a strong emphasis on spatiotemporal fluid dynamics and multi-physics foundation models.

Education

University of Illinois Urbana-Champaign, US

Aug 2022 –

PhD student in Mechanical Science and Engineering

Focus: Generative AI, Foundation Models, and Representation Learning in PDEs

- GPA: 3.64 / 4.0
- **Coursework:** Graph Machine Learning, Diffusion Flow Matching Models, Statistical Learning Theory

Pohang University of Science and Technology (POSTECH), South Korea

Mar 2016 – June 2020

B.S. in Mechanical Engineering - Summa Cum Laude (Rank : 1)

- GPA: 4.02 / 4.3 (3.89 / 4.0)

Stanford University, CA, US

June 2019 – Aug 2019

International Honors Student

- GPA: 3.914 / 4.0

Publications

Independent Latents, Robust Neural Operators

Dec 2026

J Yoo, K Kobayashi, **Jaewan Park**, S puppala, Souvik Chakraborty, SB Alam

[NeurIPS 2026](#) [🔗](#)

Stabilizing Autoregressive PDE Foundation Model Rollouts with Event-Triggered Context Healing

Nov 2026

Jaewan Park, J Yoo, K Kobayashi, S Koric, SB Alam, I Jasiuk, S Chakraborty, D Abueidda

[SC26 proceedings](#) [🔗](#)

Virtual sensing to enable real-time monitoring of inaccessible locations & unmeasurable parameters

Sep 2026

K Kobayashi, F Ahmed, **Jaewan Park**, S Sarkar, S Koric, S Chakraborty, SB Alam

[Nature Communications](#) [🔗](#)

Bridging sequential deep operator network and video diffusion: Residual refinement of spatio-temporal pde solutions

Jul 2026

Jaewan Park, F Ahmed, K Kobayashi, S Koric, SB Alam, I Jasiuk, D Abueidda

[arxiv:2507.06133](#) [🔗](#)

Under Review

Network architecture follows coupling in multiphysics systems: single vs. multiple branches in DeepONet and S-DeepONet

Jan 2026

Jaewan Park, K Kobayashi, Q Liu, S Koric, D Abueidda, SB Alam

[Communications Engineering](#) [🔗](#)

Sequential deep operator neural networks for plastic and thermo-viscoplastic transient material behavior

Aug 2025

Jaewan Park, K Kobayashi, Q Liu, AA Koric, D Abueidda SB Alam, S Koric
[International Journal of Plasticity](#)

Nonlinear Inverse Design of Mechanical Multi-Material Metamaterials Enabled by Video Denoising Diffusion and Structure Identifier

Jan 2025

Jaewan Park, S Kushwaha, J He, S Koric, Q Liu, I Jasiuk, D Abueidda
[Engineering Applications of Artificial Intelligence](#)

Predictions of transient vector solution fields with sequential deep operator network

Aug 2024

J He, S Kushwaha, **Jaewan Park**, S Koric, D Abueidda, I Jasiuk
[Acta Mechanica](#)

Advanced deep operator networks to predict multiphysics solution fields in materials processing and additive manufacturing

May 2024

S Kushwaha*, **Jaewan Park***, S Koric, J He, I Jasiuk, D Abueidda
* Co-first authors.

[Additive Manufacturing](#)

Sequential deep operator networks (s-deeponet) for predicting full-field solutions under time-dependent loads

Jan 2024

J He, S Kushwaha, **Jaewan Park**, S Koric, D Abueidda, I Jasiuk
[Engineering Applications of Artificial Intelligence](#)

Novel DeepONet architecture to predict stresses in elastoplastic structures with variable complex geometries and loads

Oct 2023

J He, S Koric, S Kushwaha, **Jaewan Park**, D Abueidda, I Jasiuk
[Computer Methods in Applied Mechanics and Engineering](#)

Ongoing Projects

Agentic reinforcement learning for automated PDE code generation

In preparation

- Training an LLM agent and shaped outcome rewards to synthesize executable finite-element solver code directly from PDE specifications.
- Generated new data for train & evaluation for multiple types of libraries (FEM, FVM, Spectral, Discontinuous Galerkin), established a failure-clustering analysis pipeline.

Benchmarking and Out-of-Distribution (OOD) Generalization for Fluid Dynamics Foundation Models

In preparation

- Developing a comprehensive benchmarking framework to evaluate the trustworthiness and generalization of PDE foundation models on large-scale Computational Fluid Dynamics (CFD) datasets.
- Formulating rigorous OOD definitions and physical validation metrics to stress-test model robustness across unseen spatiotemporal fluid regimes.

Generative Style Transfer for Histopathology with Retrieval-Augmented Flow Matching

In preparation

- Unpaired H&E-to-IHC translation via Vision Language Model and Optimal using Histopathology Foundation Models, achieving high-fidelity generation with semantic consistency.

Talks & Posters

SC26

Chicago, IL

Stabilizing Autoregressive PDE Foundation Model Rollouts with Event-Triggered Context Healing

Nov 2026

NCSA student research conference

Urbana-Champaign, IL

Transformer based neural operators for real-time accurate prediction

Apr 2026

NCSA student research conference
Multi-material lattice design with diffusion models

Urbana-Champaign, IL
Apr 2025

NCSA student research conference
Multiphysics with operator networks: Towards Additive Manufacturing

Urbana-Champaign, IL
Apr 2024

Technical Skills & Systems (Selected)

ML & Optimization: PyTorch, Hugging Face Diffusers/Accelerate/Transformers, LoRA/QLoRA, vLLM

Scientific Computing: Operator learning, FEniCSx, Abaqus, Ansys

Experimentation: Hydra-based automation, Weights & Biases, Git

Systems & HPC: Linux, Bash, SLURM (job arrays), CUDA/cuDNN, conda/mamba, Docker/Apptainer

Language: Python, C, C++, R

Teaching & Mentoring

Teaching Assistant — TAM 210/211 (Statics)

Urbana-Champaign, IL
Jan 2023 – May 2023